

GAURAV SHARMA

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Education

Bharati Vidyapeeth University, College of Engineering

Pune, India

B.Tech in Computer Engineering — Expected Graduation July 2027

Aug 2023 – Present

- **CGPA: 8.77 / 10** Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C

ML & Data: PyTorch, Mask R-CNN, EfficientNet, scikit-learn, NumPy, Pandas, Weights & Biases, CUDA

Web & Backend: Next.js 14, FastAPI, Node.js, Express, React, Tailwind CSS, REST APIs, WebSockets

Databases: PostgreSQL (Supabase), SQL, MongoDB

Cloud & DevOps: Azure Document Intelligence, Vercel, Docker, GitHub Actions (CI/CD), Git, Plotly, h5py

Research

Tooth-AI: Automated Dental Anomaly Detection | *PyTorch, Mask R-CNN, EfficientNet, CUDA*

2024 – 2025

- **Lead author** on a medical CV pipeline for OPG (panoramic X-ray) analysis with Prof. Nisha Auti, BVCOE; accepted at **RRRAI 2026**.
- Instance segmentation on **6,992 annotated images** using Weighted Box Fusion ensemble and FDI-standard geometric post-processing; achieved high-fidelity tooth-wise localisation across all 32 teeth.

Open Source Contributions

- **sktime & OpenML:** Implemented **MAAPE & MSLE** forecasting metrics from scratch in `sktime` — both merged through full agile PR lifecycle (code review, CI validation, maintainer sign-off); includes functional form, class wrapper, multioutput support, and CI test params. Open PR **#9295**: vectorised hierarchical evaluation loop (PERF). Proposed ARFF serialisation fix in `openml-python` (PR **#1490**).
- **Jenkins (CI/CD Infrastructure):** Diagnosed and fixed a JDK provisioning failure on macOS ARM64 in the `plugin-modernizer-tool` CI/CD pipeline — root cause: missing `Contents/Home` path in `.jdk` bundles. PR **#1381** merged; unblocked builds across ARM64 environments.
- **Others:** `gprMax` PR **#605** merged — reactive A-scan/B-scan radargram viewers with live simulation geometry rendering (marimo, Plotly); active **GSoC 2026** proposal. `mllam/neural-lam` — Sphinx-Autodoc POC for PyTorch Lightning documentation pipeline with two-tier CI/CD mock strategy for heavy ML dependencies (PR open).

Projects

FleetFusion | *Python, Pathway, Next.js, OpenAI API, WebSockets, PostgreSQL* | [GitHub](#)

1st Place – GenAIVERSE | 2025

- Real-time data pipeline ingesting live GPS streams via Pathway for sub-second fleet analytics; AI-powered contract compliance engine with live WebSocket dashboard. Won **1st place against 100+ teams** at a national AI hackathon.

GatiFlow | *React, TypeScript, Tailwind CSS, Leaflet.js, Java* | [GitHub](#)

SIH 2025 Institute Nominee | 2025

- Public-infrastructure smart city dashboard: real-time traffic maps, live camera feeds, 4-intersection signal control, weather and emergency response modules; **selected as institute nominee for SIH 2025**.
- Designed and benchmarked a **tuned adaptive signal algorithm** over **30 Monte Carlo runs** against 3 baselines (static, proportional, subtractive); consistently achieved lowest average vehicle wait time.

Experience

Computer Society of India (CSI) – BVCOE Chapter

Sep 2025 – Present

Technical Head

Pune, India

- Drive technical strategy for a **200+ member** AI/ML-focused student chapter; plan and deliver hands-on bootcamps, speaker sessions, and inter-college events in close collaboration with industry professionals.

Mallard Marketing Management

Jun 2022 – Apr 2023

Research & Web Development Intern

Dubai, UAE (Remote)

- Synthesised **1,000+ user behaviour data entries** to surface engagement trends; led WordPress build for Dholakiya Foundation improving donor sign-ups **25%** and engagement **30%** via SEO and Google Analytics.

Achievements & Leadership

Hackathons: GenAIVERSE (**1st Place, 100+ teams**) · Claude Challenge 2026 (Coding Period) · SIH 2025 (Institute Nominee & Team Leader) · Convolv 4.0 – IIT Guwahati · StatHack – IIT Goa (Kaggle) · Loop 1.0 – National Level

Leadership & Social Impact: ACES Media & Cultural Head (Aug 2025 – Present) · Led 12-member team; short film screened at IIT Bombay · Volunteer, Aashman Foundation (Jun–Sep 2022) — community outreach for underprivileged children, Jaipur

Certifications: Programming with Generative AI – **NPTEL Elite**, IISc Bangalore (Aug–Oct 2025) · DSA with Python – NPTEL, IIT Madras (Jan–Mar 2025) · AI – AcmeGrade (2024) · Python Mastery – Code with Mosh (2025)